

Pain Managemnt and Regional Anesthesia Course

Introduction

Acute Pain Management!



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Contents

- Objective
- Introduction
- Anatomy and physiology of pain
- Consequences of untreated Pain
- Pain Assessment
- Methods of treating acute Pain



Objective

- Define pain
- Describe pathways of pain
- Discuss about types of pain
- Discuss the negative effects of pain
- Manage perioperative pain effectively

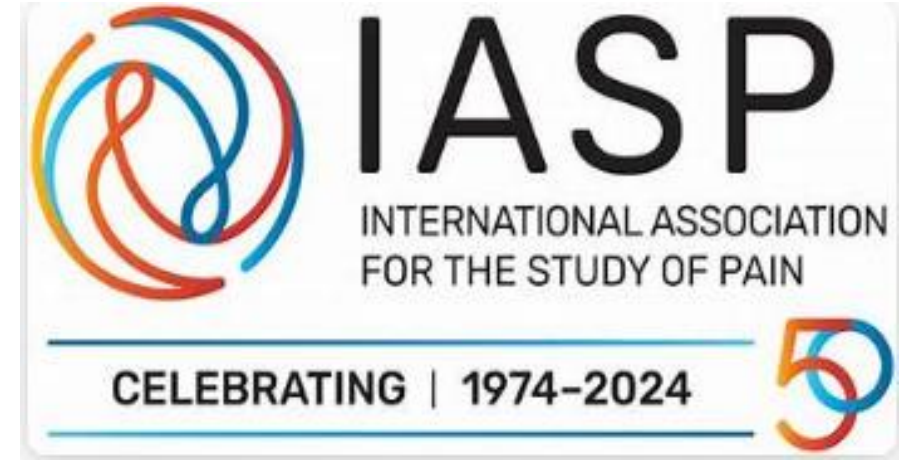


Introduction



- Pain is a vital function of the nervous system in providing the body with a warning of potential or actual injury
- It is both a sensory and emotional experience, affected by psychological factors such as past experiences, beliefs about pain, fear or anxiety and it is always a personal experience
- Nature has made sure that pain is a signal we cannot ignore
- Pain is a protective modality.

Definitions



- Latin word 'poena', means punishment or penalty.
- IASP 2020 pain definition

"Pain is an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage."

Definition continued...

- **Nociceptors**

- Are free (bare) nerve endings found in the skin, muscle, joints, bone and viscera which are sensitive to noxious stimuli

- **Noxious stimuli**

- are stimuli that elicit tissue damage and activate nociceptors.

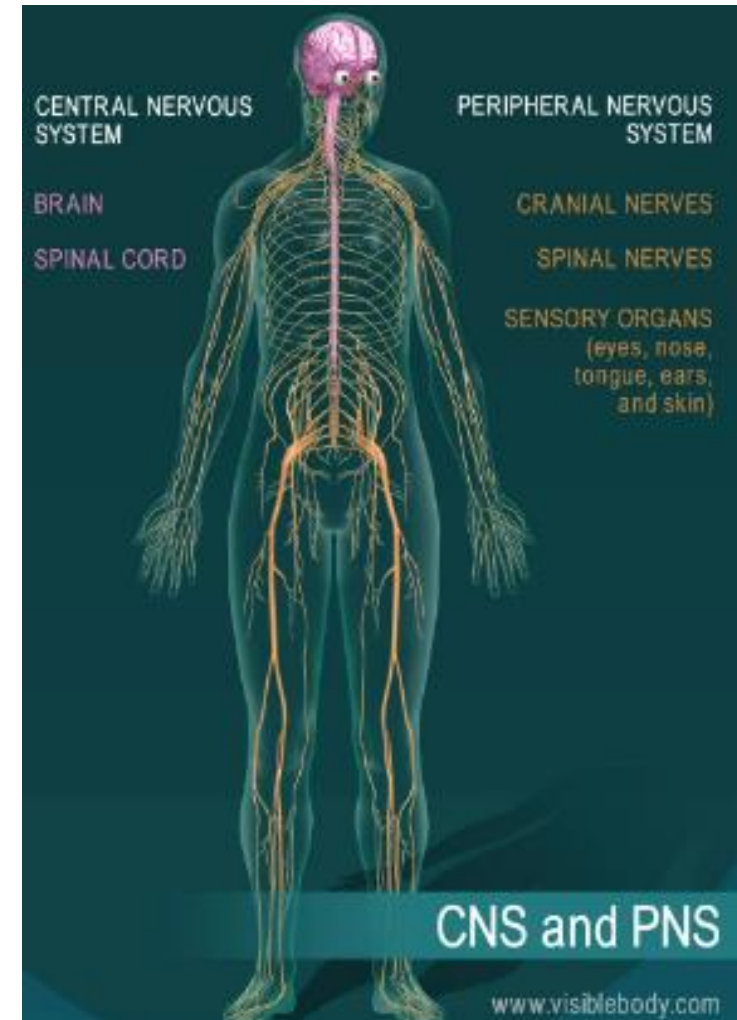
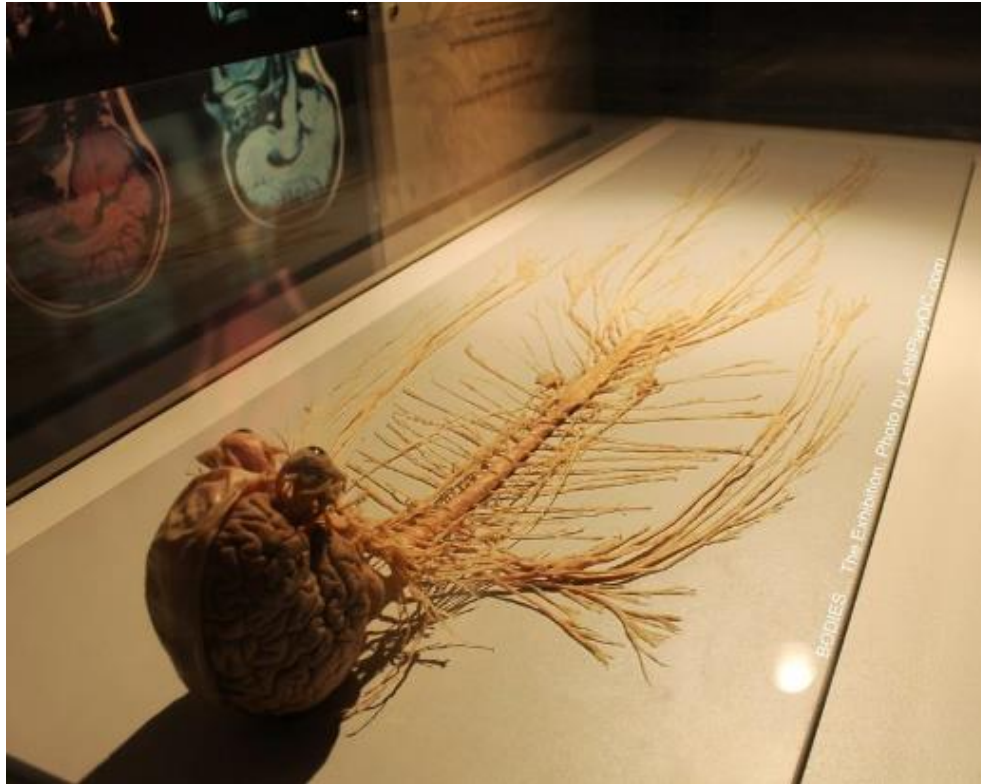
- **Inflammatory mediators**

- are chemicals released from damaged tissue and can stimulate nociceptors directly and also act to reduce the activation threshold of nociceptors (primary sensitization).

Anatomy and physiology of pain

Pain pathway

- It is a complex system that transmits signal from the site of injury through the nervous system to the brain where these signals are interpreted as pain.
- Human Nervous System



Pain pathway

- Components of pain pathway

- Nociceptors

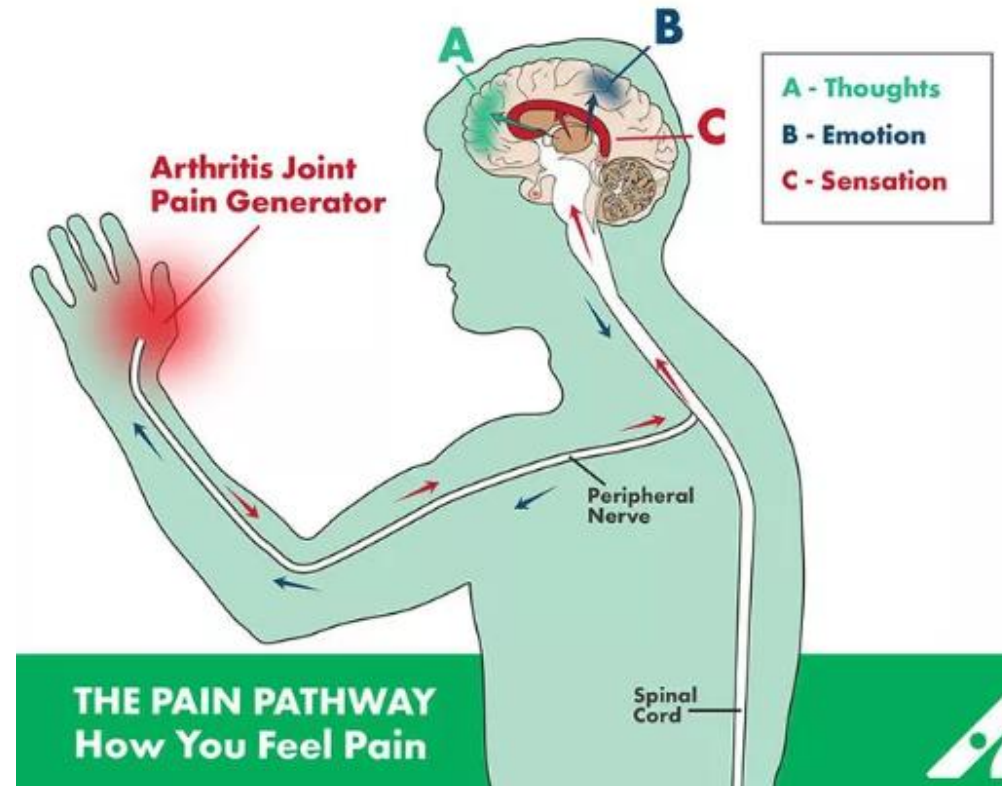
- Neural tissues

- Peripheral Nervous System

- Central Nervous System

- Autonomic Nervous System

- Neurotransmitters

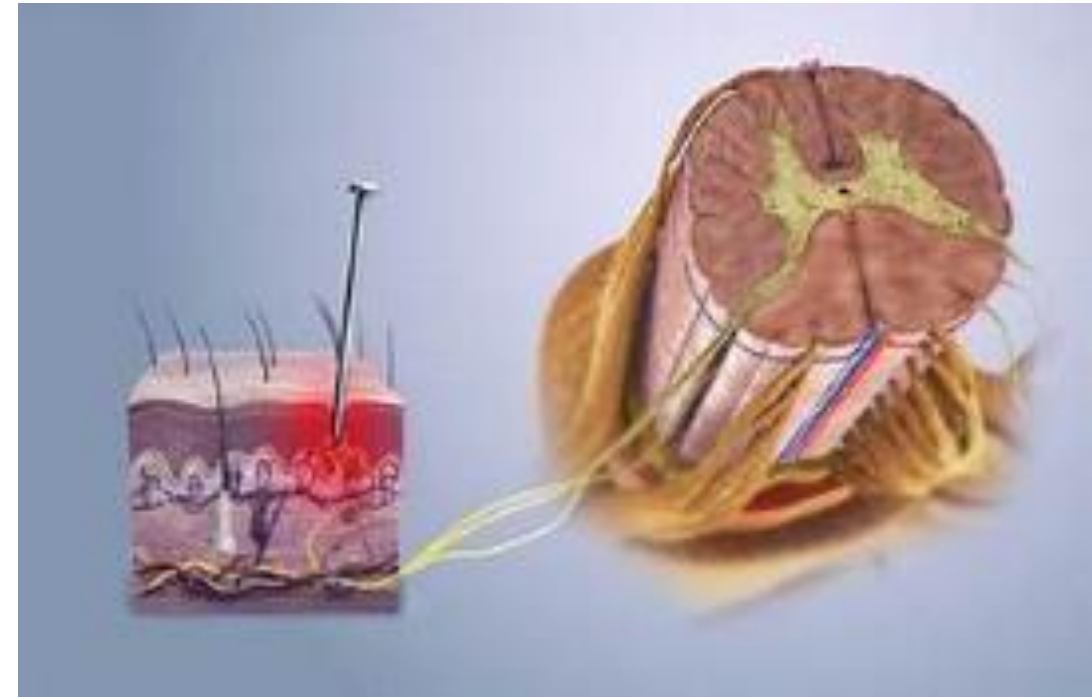


Neuronal fibers

Fibre type	Function	Diameter (μm)	Conduction (m.s^{-1})	
A α	Proprioception, motor	10–20	100	Thickly myelinated
A β	Touch, pressure	5–10	50	
A γ	Muscle spindle motor	2–5	25	
A δ	Pain, temperature, touch	2–5	25	Lightly myelinated
B (autonomic)	Preganglionic	3	10	
C	Pain, temperature	1	1	Unmyelinated
C (sympathetic)	Postganglionic	1	1	

Nociceptors

- Free nerve endings of A δ & C-fibers
 - A δ – “fast pain” (sharp) – can be located accurately
 - C-fibers – “slow pain” (dull & throbbing, poor localization of pain)
- Polymodal
 - Respond to variety of stimuli: mechanical, chemical, thermal
- Higher activation thresholds
 - Extremes of temperature and pressure
 - Stimuli in the noxious range i.e. potentially damaging to tissue
- Visceral
 - Respond to inflammation, distension and ischemia



Ascending Pain Pathways

Nociceptor Activated



Impulse arrives at dorsal horn of spinal cord
(First Order Neurone)

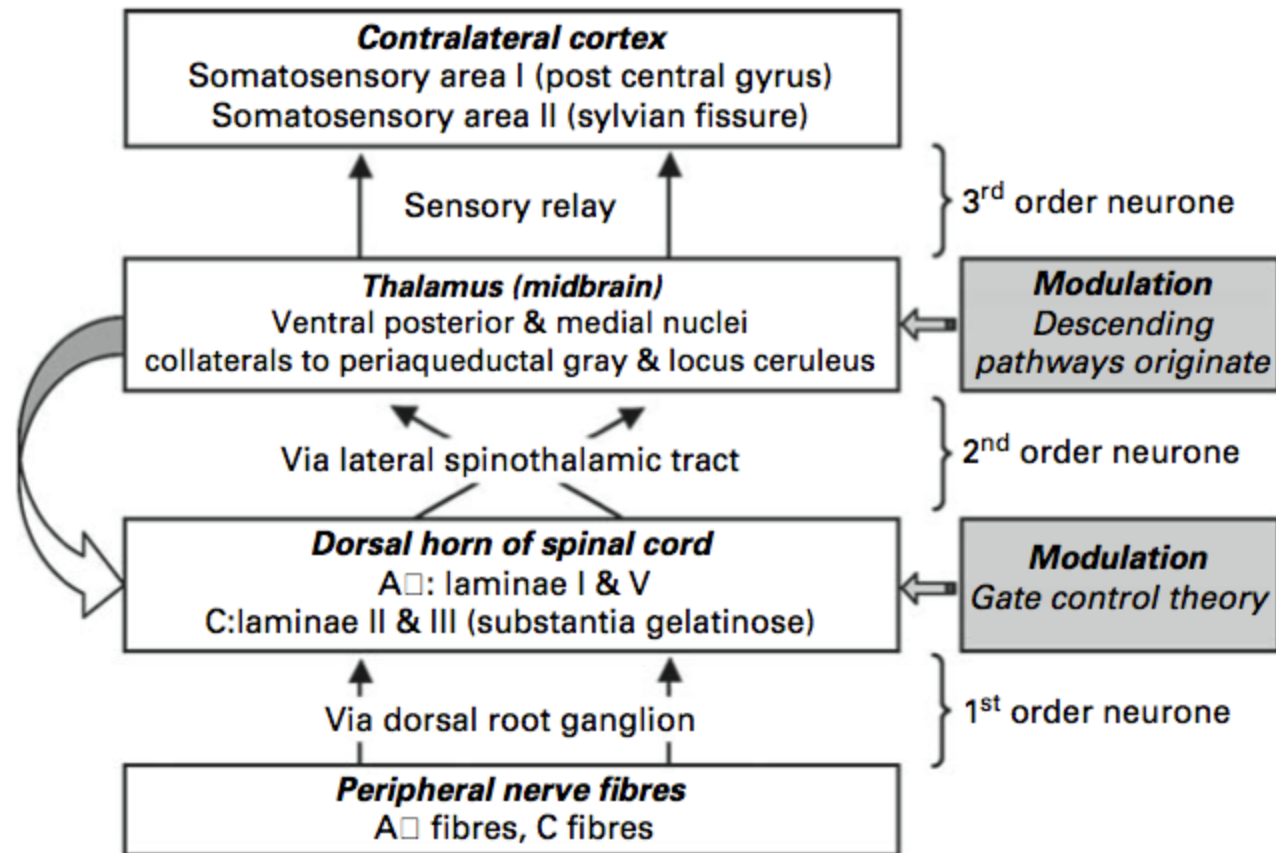


Ascends via spinothalamic tract, crossing to contralateral side
(Second Order Neurone)



Reaches thalamus and is relayed to cerebral cortex
(Third Order Neurone)

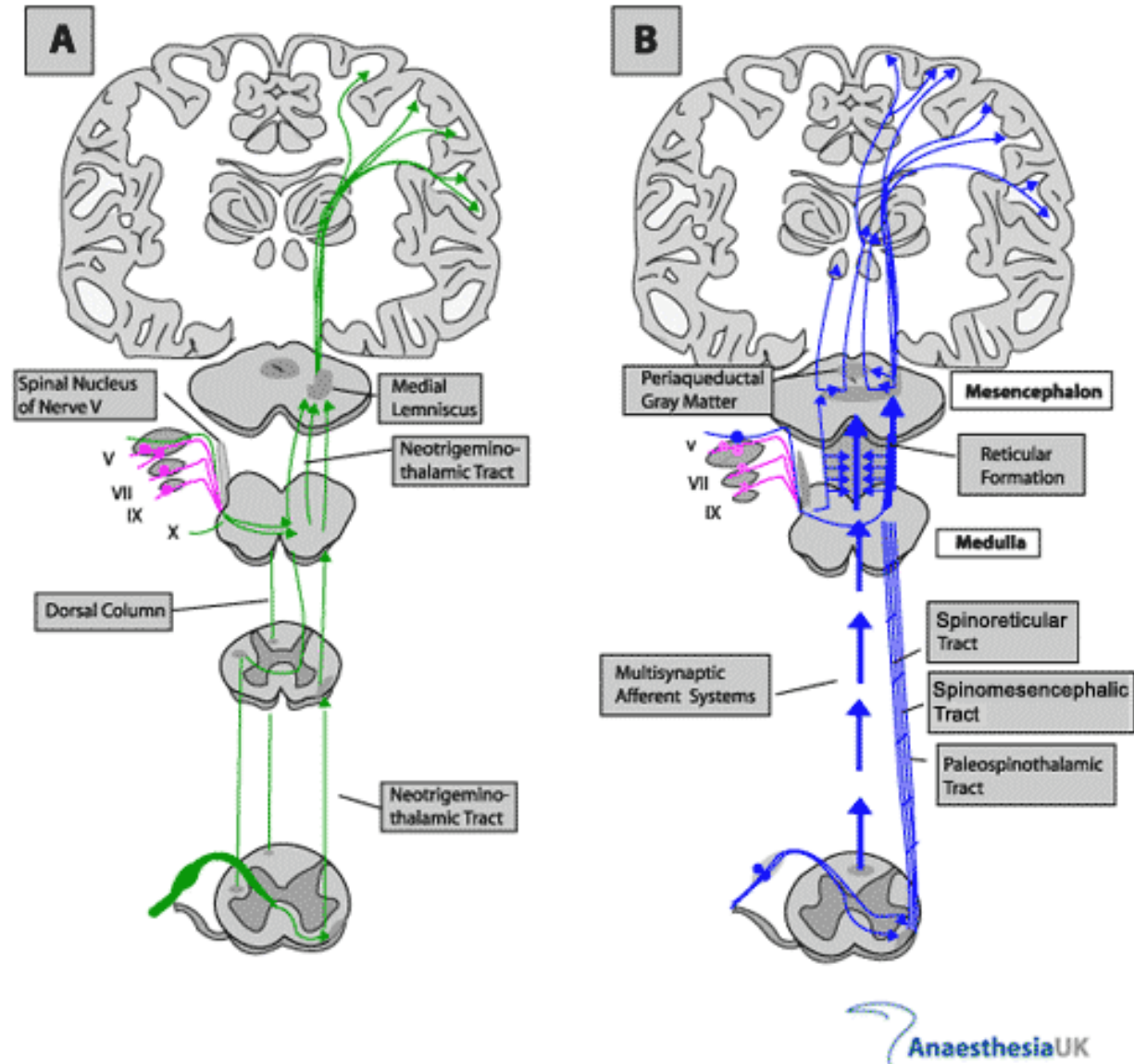
Ascending Pain Pathways



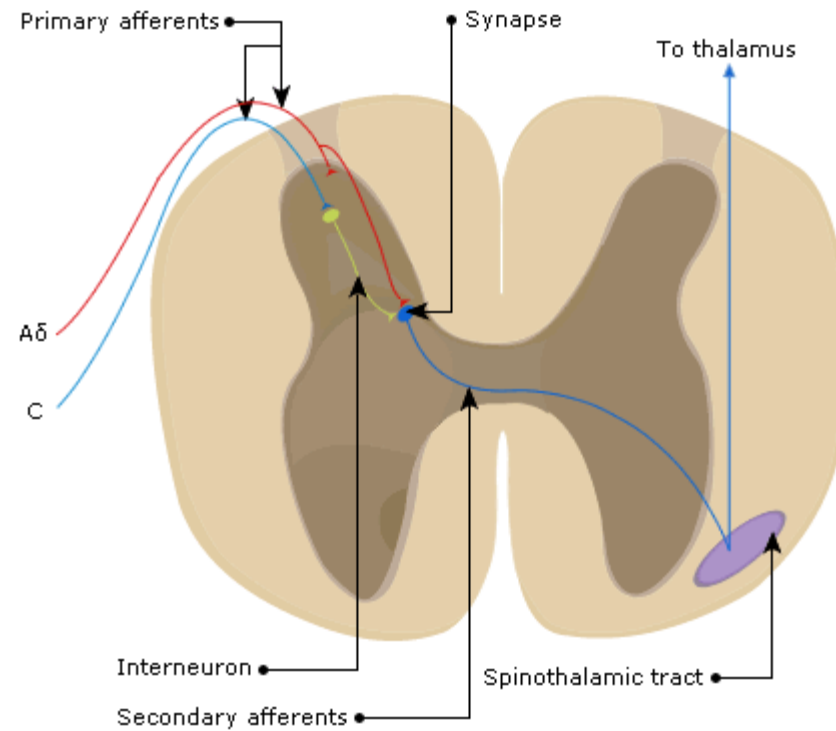
Pain pathway con'd

- In addition to **Somatosensory cortex**, pain signals will be send to:
 - **Reticular Formation** in brain steam - prevents sleep
 - **Hypothalamus** - Fight/Flight response activates: sweating, hypertension, tachycardia
 - **Amygdala** - unpleasant emotion: anxiety, fear, irritability...
- Once the pain is perceived, the descending pathway activates to dampen the pain intensity.

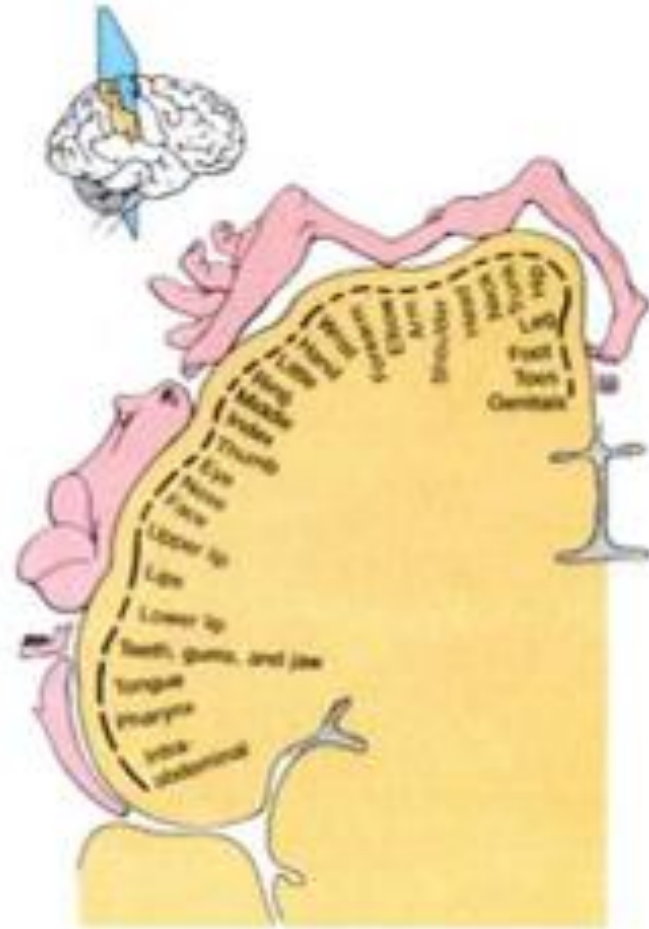
Ascending Pain Pathways



Primary & Secondary Afferents



Higher Cortical Projections



(a) Somatosensory cortex in right cerebral hemisphere

Homunculus



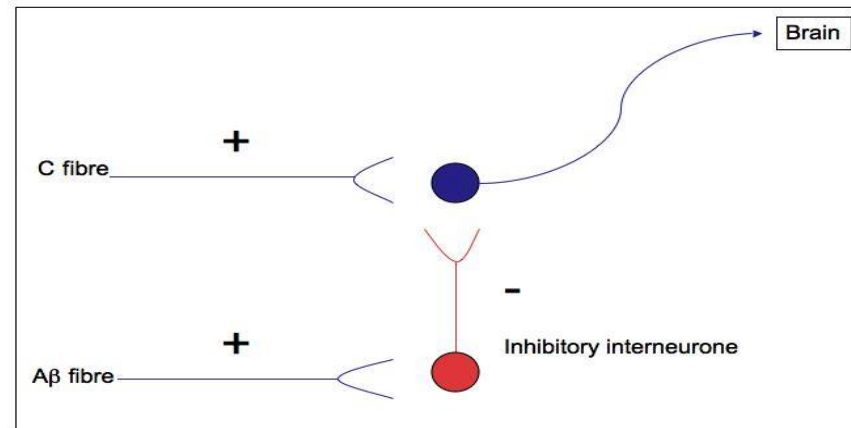
Physiology of Pain transmission

Gate control theory of pain

- Ronald Melzack and Patrick Wall in 1965
- There is a 'gate' mechanism in the CNS that opened to allow pain messages through to the brain and closed to prevent them getting through.

The gate control theory...

- Stimulation of A β fibers activates inhibitory interneurons in the dorsal horn



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Sensory Adaptation

- Most senses habituate to ongoing stimuli
 - Unpleasant smells
 - Loud music
- Pain is unusual in this respect
 - Repeated / ongoing noxious stimuli result in enhanced perception of pain (hyperalgesia)

Types of pain

- Based on their cause
 - Nociceptive pain
 - Neuropathic pain
 - Psychogenic pain
- Based on duration:
 - Acute
 - Chronic

Nociceptive pain

- Nociceptive pain is caused by **any injury to body tissues**, e.g, a cut, burn or fracture
- It is due to the stimulation of nerve fibers that transmit signals in a **normal** way from nerve endings to brain centers

Nociceptive pain...

- **Somatic pain:** pain originating from muscle, soft tissue or bone.
 - It is usually well localized and described as sharp, aching, or burning.
 - It may be worse with movement
- **Visceral pain** – pain originating from internal organs or viscera surrounding them.
 - It is usually less well localized and can be referred.
 - Often described as deep aching, cramping, or squeezing.
 - Some examples are bowel obstruction, organ ischemia, and appendicitis

Neuropathic pain

- This is caused by abnormalities in the system that carries and interprets pain
 - the problem may be in the nerves, spinal cord or brain
- Felt as a burning, tingling, shooting or electric sensation
- They may feel pain from stimuli that are not normally painful, such as light touch or cold

Neuropathic pain...

- They can also be more sensitive than normal to stimuli that are usually painful.
- Some examples are post-herpetic neuralgia, spinal cord compression, diabetic neuropathy, phantom limb, or central pain from a stroke

Psychogenic pain

- This type of pain is caused or worsened by psychological factors
- Often the pain has a physical cause, but the degree of pain and disability are out of proportion to what would be experienced by most people with a similar disorder
- This does not mean that the pain is not real, even if a physical cause cannot be found
- Any kind of pain can be complicated by psychological factors.

Acute pain

- This is short-lived pain warning the body that damage is occurring.
- It is a symptom of injury or disease at the tissue level, and tends to resolve as the injury or disease does



Acute Pain...

- Noxious
 - Short lived, e.g. following knife cut
 - Primarily transmitted by A δ fibers
- Inflammatory
 - Produces area of increased sensitivity on the injured tissue>>> **primary hyperalgesia**
 - Nociceptor sensitization by inflammatory mediators, firing threshold decreases>>> **peripheral sensitization (Secondary hyperalgesia)**

Chronic pain

- Pain that lasts >3 months, or >1 month after tissue damage healed
- Also called persistent pain, can be caused by ongoing tissue damage, such as in osteoarthritis.
- However, in some cases no physical cause for the pain can be found or pain persists long after the injury has healed.
- In many cases chronic pain is a disorder in itself rather than being the symptom of a disease process

Chronic pain ...

- Several processes can contribute to pain becoming chronic.
 - Pain receptors and neurones along the pain pathway may become too easily activated.
 - Connections between the neurons in the pathway can be altered.
 - After nerve injury, nerves may regrow but function abnormally...
- Chronic pain can persist for months or even years after an initial injury and can be difficult to treat.
- People with chronic pain may experience sleeplessness, anxiety and depression

Acute Vs Chronic pain

Acute

- Has a biological reason
- Short lived
- Sharp
- Can be treated
- Good treatment will reduce incidence of Chronic pain

Chronic

- A disease/Syndromes
- More than 3 months
- V hard to treat
- Psychological overlay
- Different symptoms
- Disability



Group Discussion: 10 minutes

- What are the consequences of untreated pain on:
 - CVS (G1)
 - Respiratory system (G2)
 - Neurologic System (G3)
 - Gastrointestinal system (G4)
 - Renal and Metabolic system (G5)

Pain: Negative Effects

■ CVS

- Increased oxygen demand of myocardium
- Lowered oxygen delivery to myocardium

■ Respiratory

- Tachypnoea
- Diaphragmatic splinting
- Decreased tidal volume
- Inability to cough

Secondary complications:

- ❖ Pneumonia
- ❖ Lung collapse
- ❖ Pulmonary Embolism
- ❖ Reduced O₂ supply to tissues

Pain: Negative Effects

- Neurological
 - ❖ Facilitation of nociceptive pathways
 - ❖ Nausea & Vomiting
 - ❖ Sleep disturbance
 - ❖ Anxiety, fear
- Gastro-intestinal:
 - ❖ Reduced gastro-intestinal motility (Ileus)
 - ❖ Delayed gastric emptying
 - ❖ Nausea & Vomiting
 - ❖ Constipation

Pain: Negative Effects

- Catabolic state and synthesis of acute phase proteins
 - increased glucose
 - Immunosuppression (danger of infection)
- Excessive stress hormone release
- Hyper-coagulability
 - DVT
- **Renal.** ADH raised. Retention
- Clotting: hypercoagulability
 - DVT
- Skin: peripheral vasoconstriction,...

Bad stuff

- Myocardial ischemia, Rhythm disturbances.
- Atelectasis. Hypoxia. Pneumonia.
- Aspiration. No food.
- Infection (glucose).
- Oedema. Confusion.
- DVT . PE
- Increased hospital stay
- Etc etc etc

Pain assessment

- Pain should be regarded as the **fifth** vital sign especially in the post operative period
- Ongoing comprehensive assessment is the foundation of effective pain management
- Assessment must determine the cause, effectiveness of treatments and impact on quality of life for the patient and their family.
- Ongoing comprehensive assessment is the foundation of effective pain management

Pain Assessment

- Accurate assessment guides pain management strategies

Remember! Pain is what the patient says it is, and exists whenever the patient says it does.

- Pain History

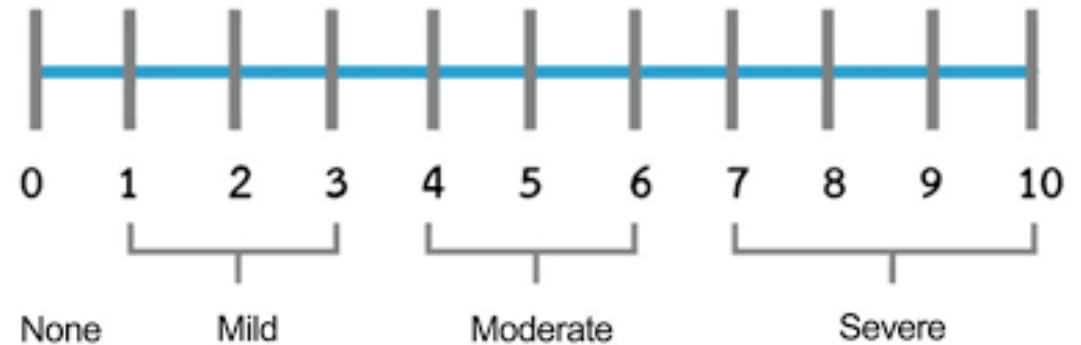
- O – Onset
- P – Provoking / Palliating factors
- Q – Quality / Quantity
- R – Radiation
- S – Severity
- T – Timing

Pain Assessment tools

- Pain assessment tools generally fall into three categories:
 - **Self-report tools:** Rely on the patient's own description of pain
 - **Observational tools:** Used when the patient cannot communicate their pain directly (e.g., infants, non-verbal adults)
 - **Physiological tools:** Monitor physiological markers associated with pain (particularly useful in patients unable to communicate verbally).

Commonly Used Pain Assessment Tools

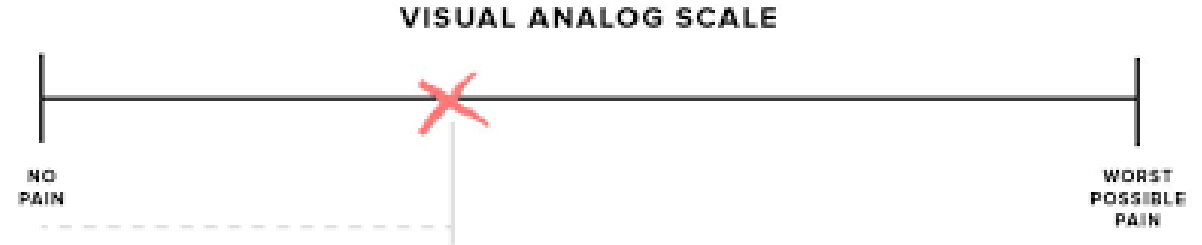
- **Numerical Rating Scale (NRS)**



- Description: Patients rate their pain on a scale from 0 (no pain) to 10 (worst possible pain)
- Best for: Adults with intact cognitive function
- Pros: Simple, universally understood
- Cons: The concept may be challenging for some

Pain Assessment Tools...

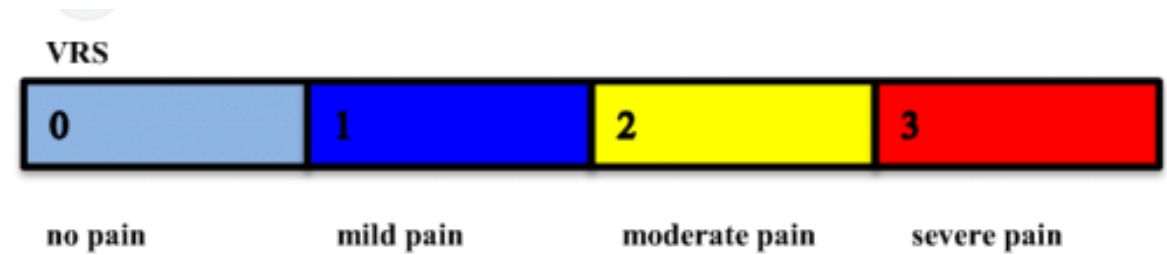
- **Visual Analog Scale (VAS)**



- Description: Patients mark a point on a 10 cm line representing their pain level
- Best for: Adults and older children
- Pros: Offers greater accuracy compared to the NRS
- Cons: Requires some motor skill and comprehension

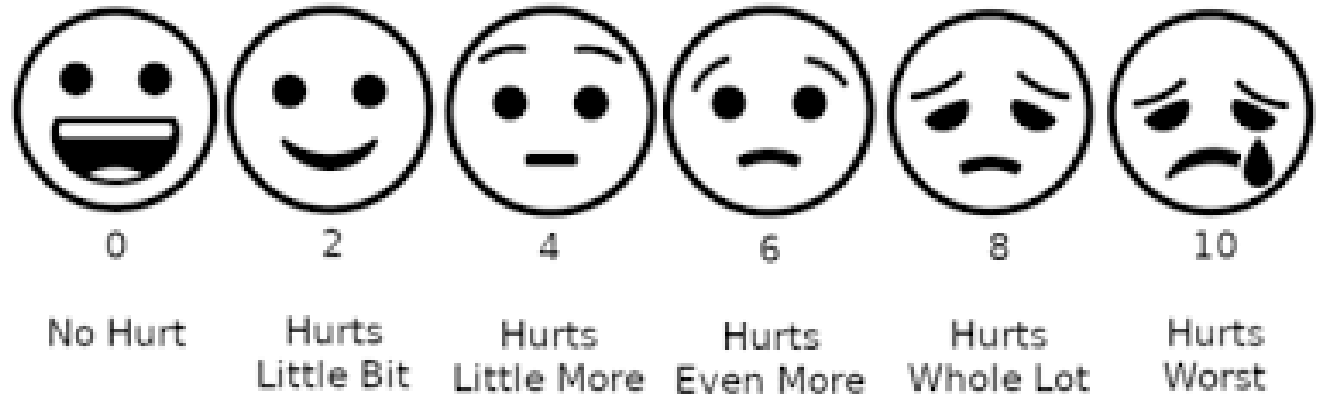
Pain Assessment Tools...

- **Verbal Descriptor Scale (VDS)**



- Description: Uses descriptive terms like "mild," "moderate," or "severe."
- Best for: Older adults and for those with limited literacy
- Pros: Easy to understand and use
- Cons: Less precise than numerical scale

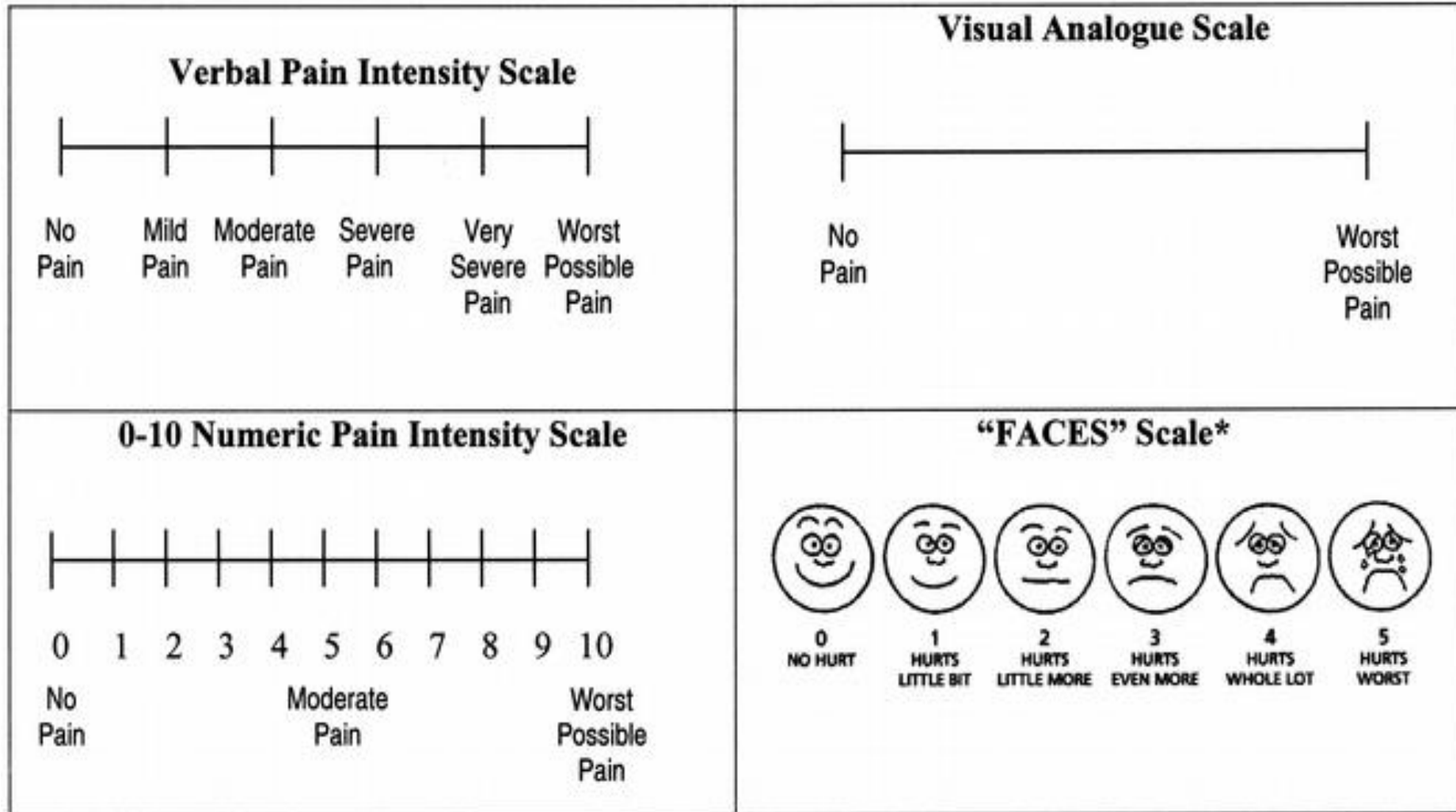
Pain Assessment Tools...



- **Faces Pain Scale - Revised (FPS-R)**

- Description: Depicts facial expressions ranging from smiling to tears
- Best for: Children (4-12 years), non-verbal adults
- Pros: Intuitive and widely applicable
- Cons: May not align perfectly with a child's personal expression of pain

Pain Assessment tools...



Managing acute pain

Methods to Treat Pain

- Pharmacologic
 - Medications (PCM, NSAIDs, Opioids, NMDA antagonists, Alpha-2 agonists)
 - Procedures
 - Regional Anesthesia
 - LA infiltration
 - Surgical Intervention
- Non-Pharmacologic

Evidence based recommendations to treat Acute Pain



- ‘Multimodal analgesia, including neuraxial opioids and non-opioid adjuncts, is recommended for all women undergoing cesarean delivery’ (Carvalho, 2017)
- ‘Multimodal approach enables unrestricted patient mobility, minimal side effects, rapid recovery, and early discharge’ (Sutton, 2017).
- ‘The use of balanced analgesia has been shown to improve acute pain management and reduce the risk of chronic pain after cesarean section’
(Lavand'homme, 2006)

Multimodal Pain Management



- Pain neurobiology is a complex of dynamic interrelated systems
- Unimodal analgesia cannot be sufficient to provide optimal pain management
- Additive & synergistic effects of multiple modes will improve outcome

Multimodal Pain Management...

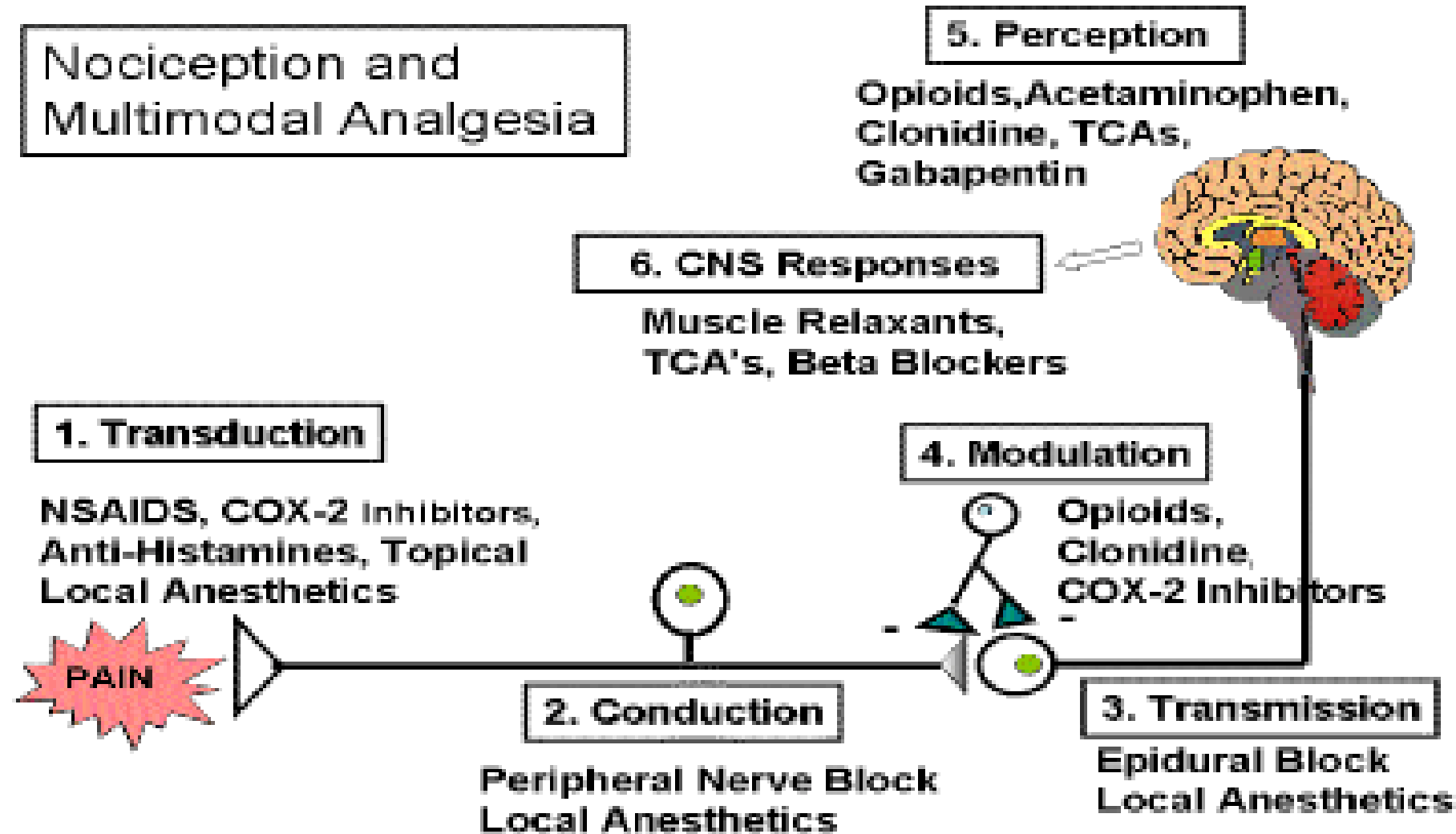
- **Definition:** MMA integrates various analgesic agents and techniques to enhance pain relief while minimizing adverse effects
- **Objective:** to achieve comprehensive pain control by targeting multiple pain pathways and mechanisms simultaneously
- **Components:**
 - Medications: Opioids, NSAIDs, acetaminophen, alpha 2 agonists
 - Regional Anesthesia: epidural, nerve blocks, infiltrations
 - Non-pharmacological interventions

MMA...

- It offers a balanced and personalized approach
- Benefits:
 - Enhanced Pain Relief
 - Reduced Opioid Dependence
 - Faster Recovery
 - Minimized Side Effects
 - Patient-Centric Care
 - Improved Patient Satisfaction
- MMA is effectively practiced using the principles of WHO analgesic ladder



Multimodal pain management Mechanisms

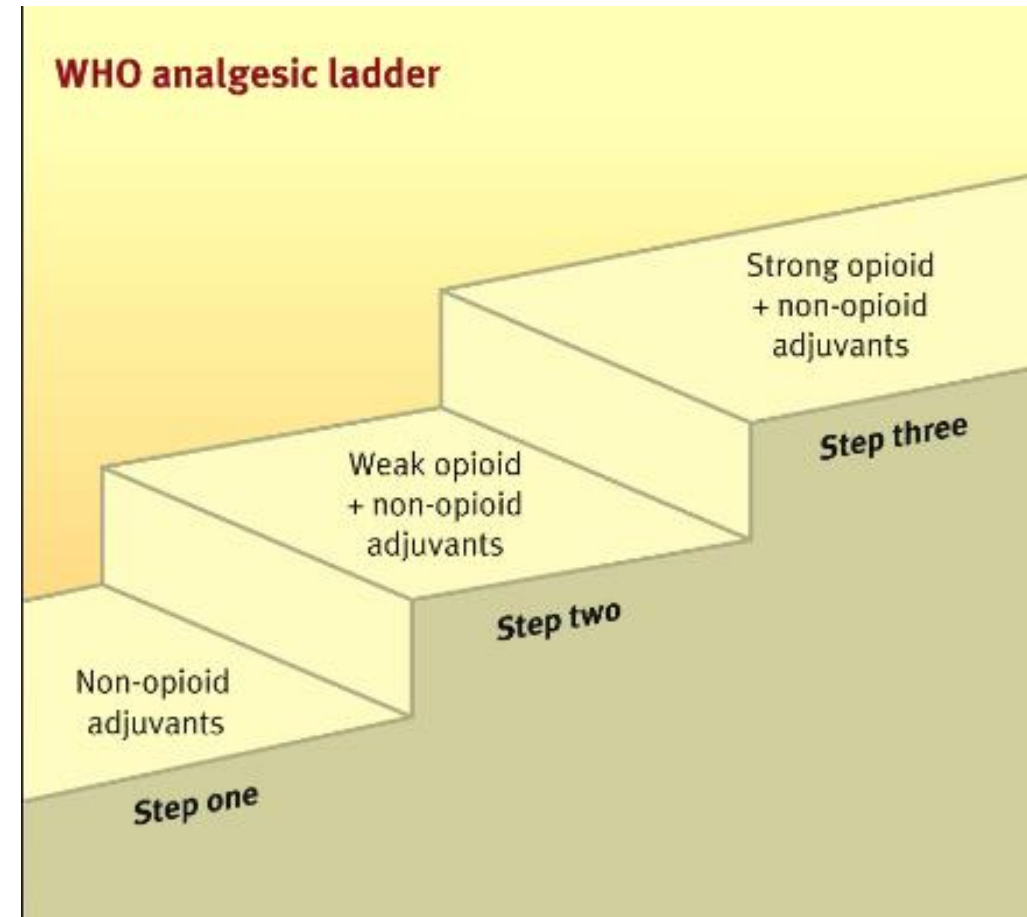


Multimodal Analgesia summary

- Using more than one drug for pain control
 - Different drugs with different mechanisms/sites of action along pain pathway
 - Each with a lower dose than if used alone
 - Can provide additive or synergistic effects
 - Provides better analgesia with less side effects (mainly opiate related S/E)

WHO Analgesic ladder

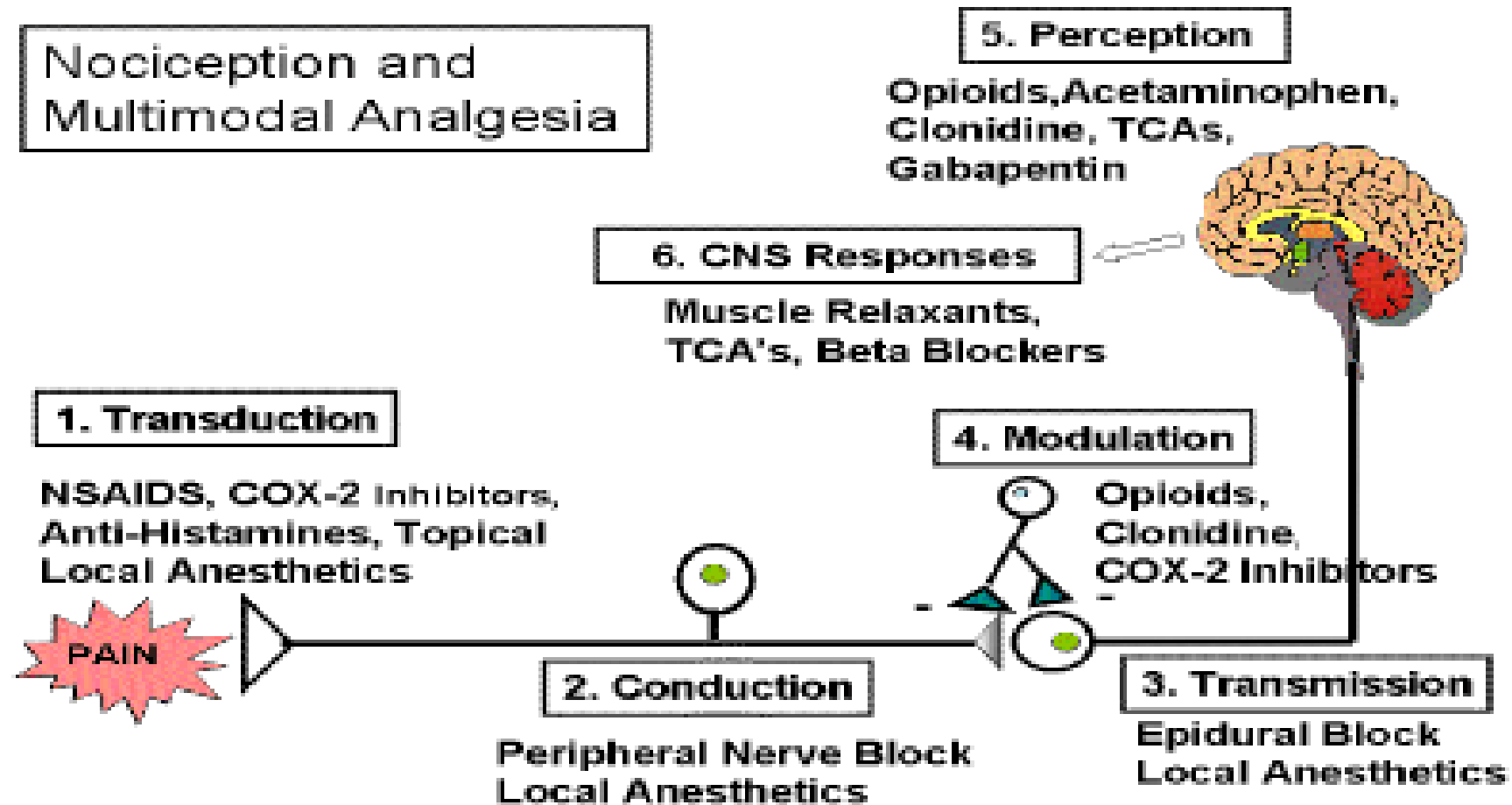
- The WHO analgesic ladder is a three-step approach to pain management
 - Step 1: Non-opioid analgesics (PCM and NSAIDs) for mild pain
 - Step 2: Weak opioids (codeine or tramadol) plus non-opioids for moderate pain
 - Step 3: Strong opioids (e.g., morphine) + non-opioids for severe pain



Advantages of WHO analgesic ladder

- Significant catalyst for improving acute and cancer pain treatment
- Legitimized the use of opioids
- Avoids the use of two drugs of the same class
- Reduces cost
- Improved pain management
- Reduced opioid side effects

Multimodal Pain Management



Summary

- Post op pain after surgery is very prevalent
- Inadequately treated post op pain has lots of consequences
- Pain assessment should be considered as a 5th vital sign in the post op period
- Multimodal pain management is the best strategy to control post op pain



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